

Math 1352 Final Exam - Spring 2011

Calculators are not allowed on this exam. Work all questions completely. Show all work as described in class. Copyright 2011 Dept of Mathematics and Statistics, Texas Tech University. Unauthorized reproduction prohibited.

1. Consider the region bounded by $y = x^2$ and $y = 2x$. **Set up** integrals to find the
 - (a) Area of the region
 - (b) Volume of the solid formed by rotating the region about the x-axis.
 - (c) Volume of the solid formed by rotating the region about the line $x = -5$.
2. For the vectors $\mathbf{u} = \langle 1, 0, -2 \rangle$ and $\mathbf{v} = \langle 0, -3, 1 \rangle$, find
 - (a) $\mathbf{u} + 3\mathbf{v}$
 - (b) $\mathbf{v} \times \mathbf{u}$
 - (c) The cosine of the angle between $\mathbf{u}$ and $\mathbf{v}$

3. Sketch the graph of $r = 3(1 - \cos \theta)$.

4. Evaluate the following integrals:

(a) $\int \frac{1}{\sqrt{x^2 + 9}} dx$

(b) $\int x \sec^2 x dx$

(c) $\int \frac{5x}{(x-1)(x+4)} dx$

(d) $\int \cos^2(3x) dx$

5. Does the series $\sum_{k=1}^{\infty} \frac{3^k}{4^k}$ converge? If so, find the sum. If not, explain why.

6. Does the sequence $\left\{ \frac{3^k}{4^k} \right\}$ converge? If so, find the limit. If not, explain why.

7. Do the following series converge or diverge? Identify any tests you use and show all the work needed to apply them.

(a) $\sum_{k=1}^{\infty} \frac{(2k)!}{3^k}$

(b) $\sum_{k=2}^{\infty} \frac{\cos k}{k^2}$

(c) $\sum_{k=2}^{\infty} \frac{3k - \sqrt{k}}{k^4 + 1}$

8. Find the first 3 terms of the Maclaurin series for $\ln(x + 2)$.

9. A (downward pointing) conical tank has a height of 10 ft and radius of 3 ft. It is filled with milk. **Set up** an integral to find the work done in pumping the milk out of the tank through a pipe that extends 5 ft above the top of the tank. (Note the density of milk is 64.5 lb/ft³.)